JSS SCIENCE AND TECHNOLOGY UNIVERSITY

JSS Technical Institutions Campus, Mysuru – 570006

"Fake news detector"

Mini project report submitted in partial fulfillment of curriculum prescribed for the Web Technology (22CS620) course for the award of the degree of

BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the work entitled **“Fake news detector”** is a bonified work carried out **by Nitish K, Deepak L, Karthik M S, and M Arun kumar** in partial fulfillment of the award of the degree of **Bachelor of Engineering in Computer Science and Engineering of SJCE, JSS Science and Technology, Mysuru during the year 2026**. It is certified that all corrections / suggestions indicated during CIE have been incorporated in the report. The mini project report has been approved as it satisfies the academic requirements in respect of mini project work prescribed for the Machine Learning (22CS610) course.

Course In Charge and Guide

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ABSTRACT

Fake news has become a major problem in today's digital world due to the rapid growth of social media and online news platforms. False information spreads quickly and influences public opinion, causing confusion and misinformation. Manually verifying every news article is difficult and time-consuming. Therefore, an automated system is required to detect fake news efficiently.

This project presents a Fake News Detection System using Machine Learning and Natural Language Processing (NLP) techniques. The system analyzes news articles and classifies them as fake or real. The dataset used for this project is collected from Kaggle and contains fake and real news articles.

The project includes data preprocessing, TF-IDF vectorization, feature extraction, model training, prediction, and evaluation. Logistic Regression is used as the primary classification algorithm because of its high accuracy and efficient performance. The developed system helps reduce misinformation spread and improves awareness regarding fake news detection using Artificial Intelligence.

1. INTRODUCTION

Fake news refers to false or misleading information presented as genuine news. With the growth of online platforms and social media applications, fake news spreads rapidly and affects society, politics, businesses, and individuals.

Machine Learning and Natural Language Processing provide effective techniques to automatically analyze textual content and identify whether news is fake or real. This project focuses on building a machine learning model capable of detecting fake news articles based on textual features.

The system uses text preprocessing methods and TF-IDF vectorization to convert textual data into numerical format. A classification algorithm is trained using labeled news data to predict the authenticity of news articles.

2. OBJECTIVES

The main objective of this project is to develop a Machine Learning system that automatically identifies whether a news article is fake or real.

Specific Objectives

- Detect misleading or false news articles automatically
- Reduce the spread of misinformation on digital platforms
- Analyze textual content of news articles
- Convert textual data into machine-understandable format
- Train a machine learning model for classification
- Improve awareness regarding fake news detection using AI
- Evaluate model performance using machine learning metrics

3. PROBLEM STATEMENT

With the rapid growth of social media and online news platforms, fake news spreads quickly and affects public opinion. Manually verifying every news article is difficult and time-consuming. Therefore, an automated fake news detection system is required to classify news articles as fake or real efficiently.

4. SOFTWARE REQUIREMENTS

Requirement	Purpose
Python 3.x	Programming Language
Jupyter Notebook / VS Code	Development Environment
Pandas	Data Handling
NumPy	Numerical Computation
Scikit-learn	Machine Learning
NLTK	Natural Language Processing
Matplotlib	Data Visualization
Seaborn	Graphical Analysis

5. DATASET DESCRIPTION

The dataset used in this project is obtained from Kaggle.

Data set

Link:

https://www.kaggle.com/datasets/emineyetm/fake-news-detection-datasets

The dataset contains two CSV files:

- **Fake.csv** → Fake news articles
- **True.csv** → Real news articles

Dataset Features

- News Title
- News Text
- Subject/Category
- Publication Date

The dataset is used for training and testing the machine learning model.

6. METHODOLOGY

The methodology of the project consists of several stages including data collection, preprocessing, feature extraction, model training, prediction, and evaluation.

6.1 Data Collection

The fake and real news datasets are collected from Kaggle and loaded using Pandas.

6.2 Data Labeling

Labels are assigned to classify the news:

- 0 → Fake News
- 1 → Real News

6.3 Data Merging

Both datasets are combined into a single dataset and shuffled randomly to improve model training.

6.4 Data Preprocessing

Text preprocessing is performed to clean and normalize the dataset.

Preprocessing Steps

a) Convert to Lowercase

All text is converted to lowercase format.

b) Remove Punctuation

Special symbols and punctuation marks are removed.

c) Remove Numbers and Special Characters

Numbers and unwanted characters are eliminated.

d) Remove Stop words

Common English words such as “is”, “the”, and “are” are removed.

e) Tokenization

Text is split into individual words or tokens.

f) Stemming

Words are reduced to their root forms.

Example:

Playing → play

running → run

6.5 Feature Extraction using TF-IDF

Machine learning models cannot understand textual data directly. Therefore, TF-IDF vectorization is used to convert text into numerical vectors.

TF-IDF Formula

$$TF\text{-}ID(t,d) = TF(t,d) \times \log\ \left(\frac{N}{DF(t)} \right)$$

Where:

- TF(t,d) = Frequency of term in document
- DF(t) = Number of documents containing the term
- N = Total number of documents

Purpose

- Convert text into machine-readable format
- Identify important words in documents

6.6 Dataset Splitting

The dataset is divided into:

- 80% Training Data
- 20% Testing Data

6.7 Model Training

Machine learning classification algorithms are used to train the model.

Algorithms Used

- Logistic Regression
- Naive Bayes
- Random Forest
- Decision Tree

Recommended Algorithm

Logistic Regression provides better accuracy and faster performance.

6.8 Prediction

The trained model predicts whether a news article is:

- Fake News
- Real News

7. PROJECT WORKFLOW

News Dataset

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Data Preprocessing

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TF-IDF Vectorization

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Train/Test Split

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Logistic Regression Model

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Prediction

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Fake / Real Output

8. IMPLEMENTATION

Step 1: Import Libraries

Required Python libraries are imported for:

- Data handling
- NLP preprocessing
- Machine learning
- Visualization

Step 2: Load Dataset

The fake and real news datasets are loaded using Pandas.

Step 3: Preprocess Data

Text preprocessing techniques are applied to clean the dataset.

Step 4: Apply TF-IDF

Textual data is converted into numerical vectors.

Step 5: Train the Model

The Logistic Regression algorithm is trained using training data.

Step 6: Test the Model

Testing data is used to evaluate prediction accuracy.

Step 7: Display Prediction

The system predicts whether the news is fake or real.

9. RESULTS

The Logistic Regression model successfully classifies fake and real news articles with high accuracy. TF-IDF vectorization improves text representation and enhances model performance.

The system effectively predicts the authenticity of news articles using textual features.

Example Output:

News: "Government launches new policy"

Prediction: Real News

10. ADVANTAGES

- Fast fake news identification
- Reduces misinformation spread
- Saves manual verification effort
- Improves online news reliability
- Useful for social media monitoring

11. APPLICATIONS

- Social Media Platforms
- News Websites
- Online Content Verification
- Journalism and Media Monitoring
- Fact-Checking Systems